

Is the Retiree Willing to Plan for Complete Portfolio Depletion?

Traditional safe withdrawal rate literature regularly assumes that retirees will choose a withdrawal rate that will leave precisely no wealth at the end of the retirement period. Retirees cling to inflation-adjusted withdrawal amounts, which leaves them playing a game of chicken as their wealth plummets toward zero. In addition, these hypothetical retirees do not make any adjustments for the fact that as their final planned year of retirement approaches, they are increasingly likely to live longer than the age they have planned for. The classical assumptions also present retirees with no particular desire to leave a bequest, an estate, an inheritance, or whatever you may like to call it. The objective of the classical studies is to get a handle on the maximum sustainable withdrawal rate from a portfolio of volatile assets over a thirty-year retirement period without worrying about whether anything will still be left at the end. When we talk about using a safe withdrawal rate, we are describing a situation in which remaining wealth is potentially allowed to fall to zero at the end of the planning horizon.

The safe withdrawal rate approach is meant to typically provide leftover funds at the end of the time horizon when not in a worst-case scenario. But the analysis will be different if we specifically incorporate a desire that our worst cases still preserve some assets at the end of the time horizon. I aim to investigate how withdrawal rate decisions may change when retirees specifically incorporate a desire to leave a bequest, which I will summarize here as either maintaining the *nominal* value of retirement date wealth at the end of the thirtieth year, or maintaining the *real* value of retirement date wealth at the end of the thirtieth year. The value of wealth may decline and rebound in the interim, as I am only checking the value of wealth after the thirtieth year.

I am using the same assumptions as described in my discussion of William Bengen's SAFEMAX:

- withdrawals at the start of the year,